



الاختبار النصفى لمقرر التعلم العميق المتقدم (AI4204)

Midterm Exam – Advance Deep Learning (AI4203)

الفصل الدراسي	تاريخ الاختبار	مدة الاختبار
472	الأربعاء 1447/10/20 الموافق 2026/04/08	ساعة

تعليمات الاختبار:

- عدد أسئلة الاختبار ١٢ سؤالاً.
- درجة الاختبار ٣٠ درجة،
- تأكد من كتابة اسمك ورقمك الجامعي ورقم الشعبة في ورقة الإجابة .
- غير مسموح باستخدام أي مذكرات أو مراجع أثناء الاختبار (Closed book exam).
- في حال احتجت إلى مساعدة أو استفسار، ارفع يدك وانتظر المراقب للرد علي ك.
- احرص على متابعة وقت الاختبار لكي تتمكن من الإجابة على جميع الأسئلة
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Exam Instructions

- The exam consists of 12 questions.
- The total exam score is 30 marks.
- Make sure to clearly write your name, university ID number, and section number on the answer sheet.
- No notes or reference materials are allowed during the exam (closed-book exam).
- At the end of the exam, submit both the question paper and the answer sheet to the invigilator.
- If you need assistance or have a question, raise your hand and wait for the invigilator to assist you.
- Please manage your time carefully to ensure you are able to answer all required questions.

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1. What are the three main parts that make up the Transformer architecture?

Embedding
Encoder-Decoder

Activation function

?! X X you should mention "Attention"

2. What is the primary function of "Positional Encoding" in Transformers?

the primary function of positional encoding is positioning the encoder in the position where it would function best within a transformer

3. In calculating self-attention, three variables taken from database terminology are used. What are they?

Q, K, V

4. In the self-attention mechanism, what does the Query (Q) represent?

- A) The stored information retrieved as a result of the attention process
- B) The current word that is "asking" which other words are most relevant to it
- ☒ C) A label used to index and match words in the attention lookup table
- D) The scaling factor applied before the SoftMax to stabilize gradients

5. The SoftMax function in the self-attention equation converts similarity scores into probabilities that sum to 1, producing attention weights for each word.

- ☒ A) True
- B) False



6. What is the key difference between Self-Attention and Masked Self-Attention regarding access to other words?

- A) Self-Attention can only look at previous words, while Masked Self-Attention can look at all words
- ☒ B) Self-Attention allows every word to attend to all other words, while Masked Self-Attention restricts each word to attend only to previous words
- C) Both allow full access to all words, but Masked Self-Attention applies dropout for regularization
- D) Masked Self-Attention attends to future words only, while Self-Attention attends to past words only

7. Why do we transpose the Key matrix (K) when multiplying it with the Query

- A) To normalize the values and prevent large numbers
- ☒ B) To make the matrix dimensions compatible for multiplication
- C) To reduce the size of the matrix for faster computation
- D) To apply positional encoding to the input

8. What is the title of the seminal 2017 paper that first introduced transformer architecture, and what specific task was the model originally designed to perform?

فن الإصطناع
صوابه
the title represented a new era in deep learning as transformers revolutionized the industry and it was designed to make models more efficient and accurate.

9. In Encoder-Decoder Attention, where do Q, K, V come from?

Query key value
أين الجواب؟

Q, K, V أنت فتي لا تفهم، قل لي

10. Mention two practical uses for the "Context Aware Embeddings" produced by Encoder-Only Transformers?

يا ربي الغاي انت جالس تقدم شرح عام عن أهمية ال context
Context Aware Embeddings are used mainly when we need the context to be put into consideration and when the dis regard of it would result in inaccurate results



11. Explain what Word Embedding and Positional Encoding are, and how they are combined to form the input of a Transformer model.

Word Embedding and Positional Encoding are two of the main parts that make a transformer as we need them before applying the transformer it makes it better encoded and embedded before it becomes a proper input as transformers would function better after these processes.

12. Compare Encoder-Only and Decoder-Only Transformers. Explain the type of attention used in each, their primary goals, and why ChatGPT is considered a Generative Model.

Aspect	Encoder-Only	Decoder-Only
Attention	masked self attention	self attention
Goal	disrupted irreversibility.	using more contextual data to give an only one based on the whole inputs before
Example tasks	Giving the weather forecast	Analysing a BP monitor

Why ChatGPT is generative:

classification - sentiment Analysis - NER
Next token Prediction - Text Generation

ChatGPT is a generative tool because it doesn't have ready-made responses as it can only respond after a given prompt after generating a response which can be in text, audio, or even visual as in picture/videos.

~ Good Luck ~



Questions Overview (Marks Distribution)

Q#	Question Type	Marks
Q1	Short Answer	2
Q2	Short Answer	2
Q3	Short Answer	2
Q4	MCQ	2
Q5	True/False	2
Q6	MCQ	2
Q7	MCQ	2
Q8	Short Answer	3
Q9	Short Answer	2
Q10	Short Answer	2
Q11	Essay	4
Q12	Essay	5
Total		30